

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

Gereon Feldmann, Pit Steinbach and Christoph Bannwarth

Institute of Physical Chemistry, RWTH Aachen University, 52074 Aachen, Germany

Introduction

- The Hessian matrix is a target of interest, due to its high computational cost and its versatile applications.
- Atomistic Features, based on GFN2-xTB quantities, are used for the prediction of the target: Hessian matrix elements
- The features are permutation and rotational invariant to achieve a unique mapping onto the target.
- The zero point vibrational energy is used to quantify the precision of the model.

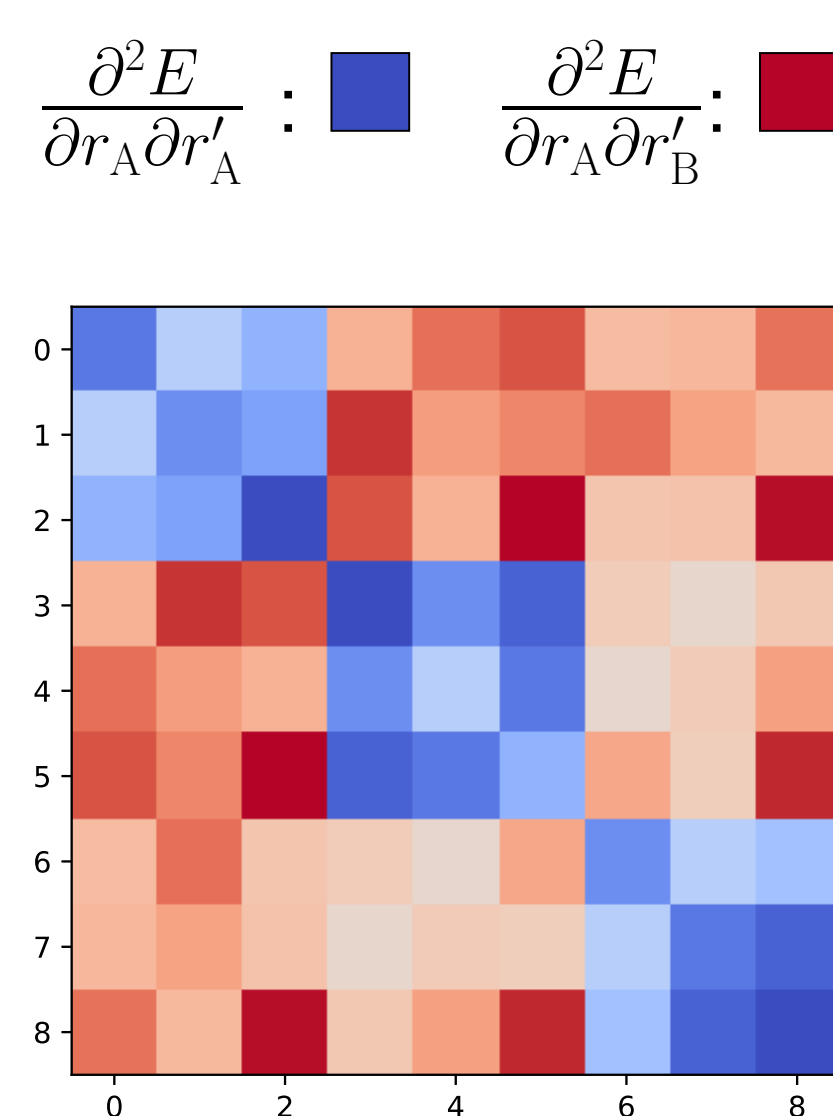
Overview of the ML Model

Origin of the Features

- Atomwise partitioned GFN2-xTB quantities are used for the features and were developed in our group
- The features are physics based and can be categorized into:
 1. Geometry: e.g. Coordination Number
 2. Density: e.g. Dipole Moment
 3. Energy: e.g. Coulomb-repulsion energy

The Hessian Matrix

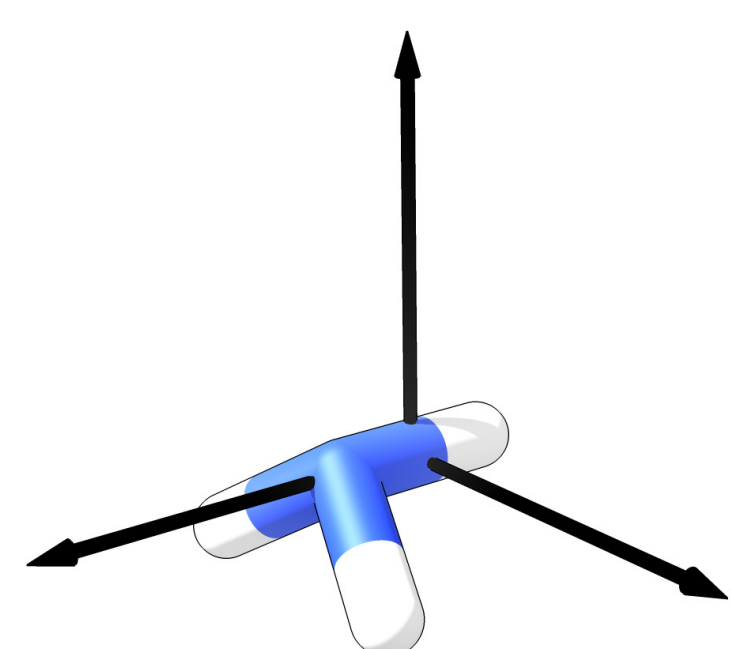
- A $3N \times 3N$ matrix of second derivatives of the energy in respect to the coordinates of the N atoms
- The derivatives are either in respect of the coordinates of a single atom or of two atoms
- Two submodels are used to describe the diagonal blocks (blue) and off-diagonal blocks (red)



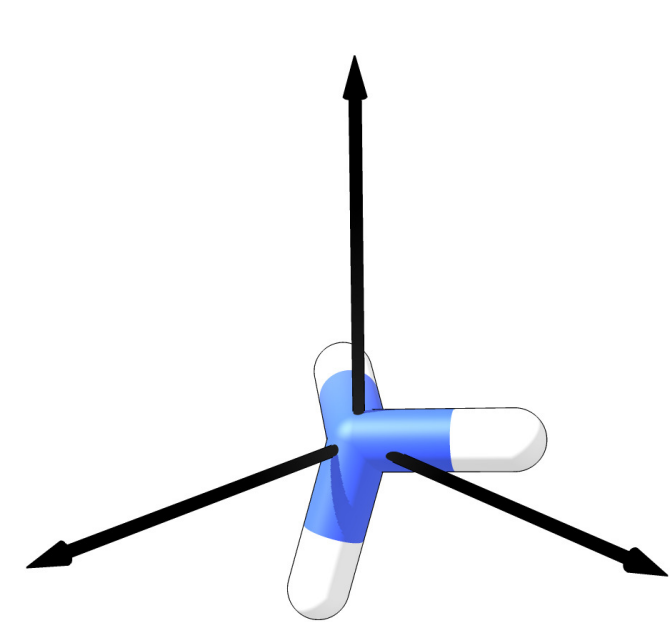
Connecting Features and Target

- The feature vector is either build up from the quantities of one or two atoms
- For ML modeling a constant feature vector length is necessary, hence two submodels are used
- The features and targets are computed for each 3×3 block in a different orientation.
- The molecular structure is oriented differently in the coordinate system in dependence of the submodel

Model ■ : Here the origin is shifted to the center of two atoms and the atoms are aligned to the z-axis.



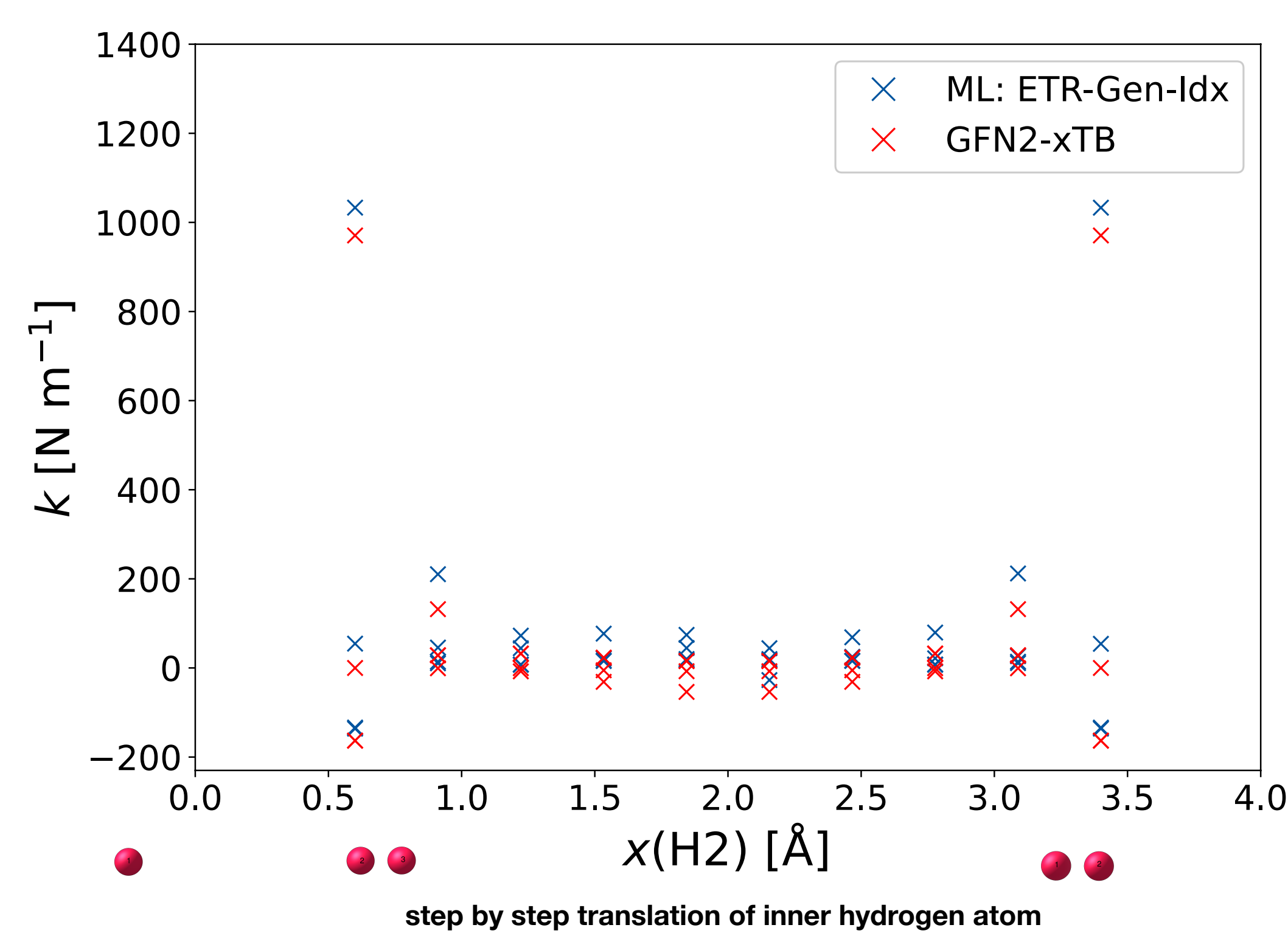
Model ■ : The origin of the coordinate system is shifted to the atom.



Unifying the Mapping of the Features

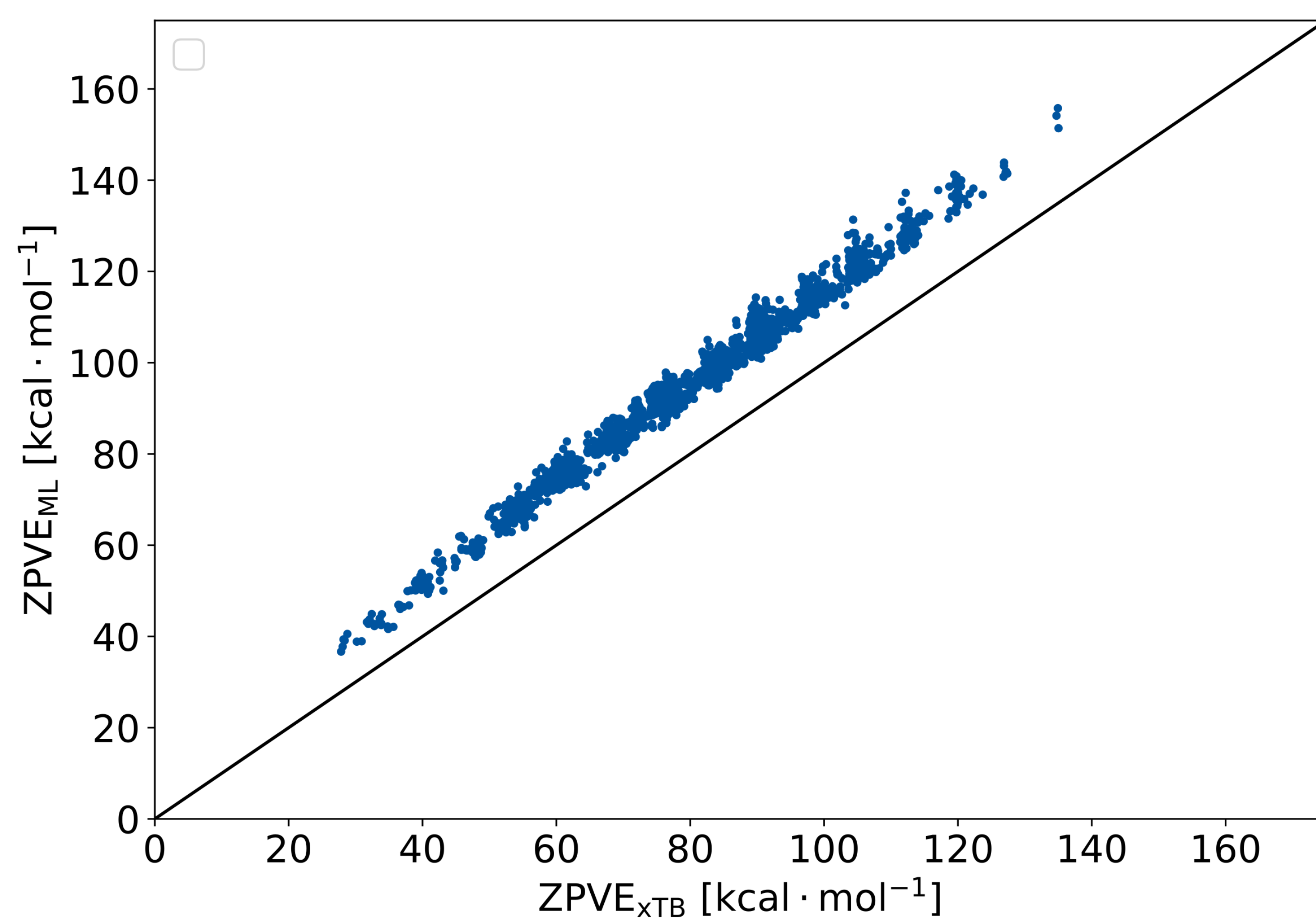
- A unique mapping of the features onto the target is achieved by different means
- An index is introduced to differ between the derivatives in respect to different coordinates of the same atom
- It is taken care of a consistent alignment to the atom oriented coordinate system

Verification of Unique Mapping



Predicted ZPVE of GBD7 Set

- Learning plots were computed using the GBD7 set, which consists of 7164 structures
- The test set was fixed to 25 % of the size of the GBD7 set
- The training set was increased up to 15% of the



Conclusion and Future Perspective

- Bullet point A
- Bullet point B
- Bullet point 3